INCH-POUND

MIL-DTL-17361/15C(SH)
19 September 2006

SUPERSEDING
MIL-C-17361/15B(SH)
1 March 1994

DETAIL SPECIFICATION SHEET

CIRCUIT BREAKERS, AIR, ELECTRIC, INSULATED HOUSING (SHIPBOARD USE) TYPES AQB-A1602 AND NQB-A1602

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-17361.

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TABLE I. Requirements.

Maximum rated continuous current	1600 amperes
Voltage and frequency	500 VAC, 60 Hz
Short-time withstand current rating	50,000 amps, 0.3 seconds
Interrupting rating	100,000 amperes, 60 Hz, AC, asymmetrical 85,000 amperes, 60 Hz, AC, symmetrical
Trip unit	Interchangeable rating plugs (See Table II)
Current limiting fuses	Not applicable
Type of connections	Back (removable mtg.)
Electric operator	Available
Undervoltage trip or shunt trip	Either or both available
Auxiliary switch	Provided as standard to protect shunt trip and remote closing solenoids when ordered: four (4) or eight (8) additional SPDT switches optionally available.
Cell switches	One SPDT switch provided standard
Handle lock	Not available
Handle boot	Not available

NOTE: NQB circuit breakers shall consist of an AQB type without trip unit and a modification kit of metallic parts.

TABLE II. Continuous current sensor ratings.

Frame rating	**AQB-A1602 sensor ratings**					
800	400	480	560	640	700	800
1600	800	900	1000	1200	1400	1600

TABLE III. Instantaneous trip settings amperes/maximum time (all 3 poles) /1.

AQB-A1602	
Frame rating	**20X**
800	16000
1600	32000

NOTE:

/1 The tolerance for each ampere setting shall not exceed ±20%.

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TABLE IV. Adjustable short-time delay pickup settings in amperes (STD) (all 3 poles) /1.

AQB-A1602								
Frame rating	**2X**	**2.5X**	**3X**	**4X**	**5X**	**6X**	**7X**	**8X**
800	1600	2000	2400	3200	4000	4800	5600	6400
1600	3200	4000	4800	6500	8000	9600	11200	12800

NOTE:

/1 The tolerance for each ampere setting shall not exceed ±10%.

TABLE V. Adjustable short-time delay settings in seconds.

AQB-A1602 trip units only			
Short-time delay "delay setting" (seconds)			
0	0.10	0.20	0.30

NOTE:

/1 The tolerance for each setting shall not exceed ±20%.

TABLE VI. Tripping characteristics – long-time delay pickup settings in amperes (all 3 poles).

AQB-A1602											
Sensor rating	400	480	560	640	700	800	900	1000	1200	1400	1600
150% of sensor rating	600	720	840	960	1050	1200	1350	1500	1800	2100	2400

NOTE:

/1 The tolerance for each ampere setting shall not exceed ±10%.

TABLE VII. Tripping characteristics – long-time delay (all 3 poles) /1 /2.

AQB-A1602	
LTD in multiples of CCS	Tripping time 60 Hz AC
6.0	4 seconds

NOTE:

/1 The tolerance for each setting shall not exceed ±20%.

/2 At other currents, delay time is inversely proportional to current squared.

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TABLE VIII. Test requirements.

<table>
<tr><th>Test</th><th colspan="3">Requirement</th></tr>
<tr><td>Mechanical endurance test</td><td colspan="3">The circuit breaker shall be subjected to 1000 close-open no-load operations.</td></tr>
<tr><td>Electrical endurance test</td><td colspan="3">The rate of operation shall be 4 operations per minute for 10 minutes and not less than one operation per minute thereafter for a total of 4000 close-open operations.</td></tr>
<tr><td rowspan="3">Temperature rise test</td><td colspan="3">Size of cable and bus bars for temperature rise tests</td></tr>
<tr><td>Copper rating-amperes</td><td>Cable size</td><td>Copper bus size (inch)</td></tr>
<tr><td>1600</td><td>Four (4) 400 MCM</td><td>Two (2) 3 x 1/4</td></tr>
<tr><td>Anti-pump test</td><td colspan="3">1. Test shall be accomplished immediately following the trip-free test (see Table V of MIL-DTL-17361).
2. With the circuit breaker “closed” and the “closed switch” maintained “closed”, the circuit breaker when tripped shall not recluse.</td></tr>
<tr><td>Short-time withstand current test</td><td colspan="3">1. Test shall be accomplished immediately prior to interrupting performance testing (see Table V of MIL-DTL-17361).
2. Circuit breaker shall be subjected to rated short-time withstand current at rated voltage and for the time duration as shown in Table I.
3. The short-time withstand current is expressed in root mean square (rms) symmetrical amperes and is measured from the envelope of the available current wave at 0.5 cycles after current is established. For three-phase circuits, the symmetrical current value shall be the average of the phase currents.
4. Circuit breaker shall perform their short-time current duty cycle with all degrees of current asymmetry produced by three-phase circuits, having a short-circuit lagging power factor of 0.15 lagging or less (X/R ratio of 6.6 or greater).
5. AC test circuits shall be in accordance with the test circuits for the interrupting performance test in Table V of MIL-DTL-17361.</td></tr>
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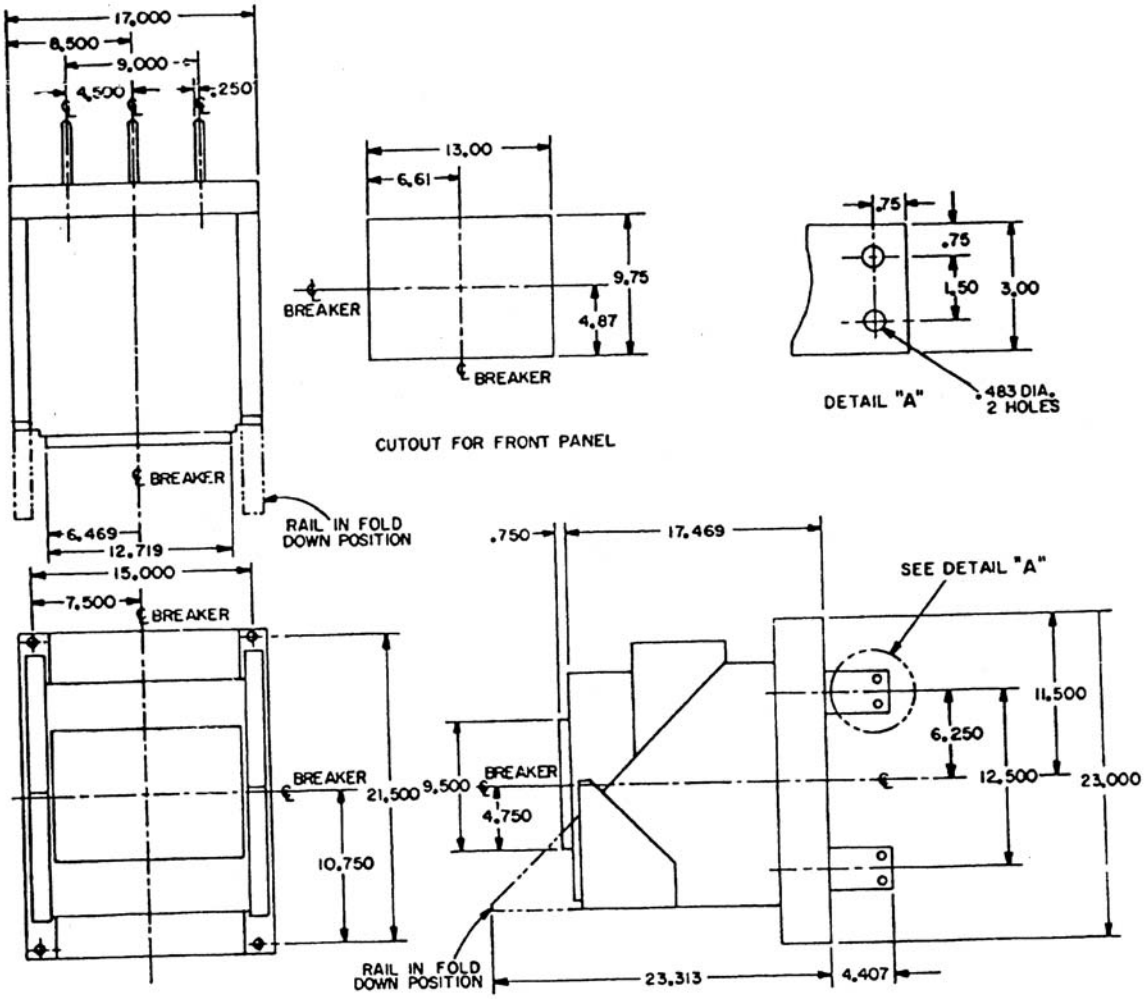


FIGURE 1. Circuit breaker types AQB-A1602 and NQB-A1602, outline and mounting dimensions.

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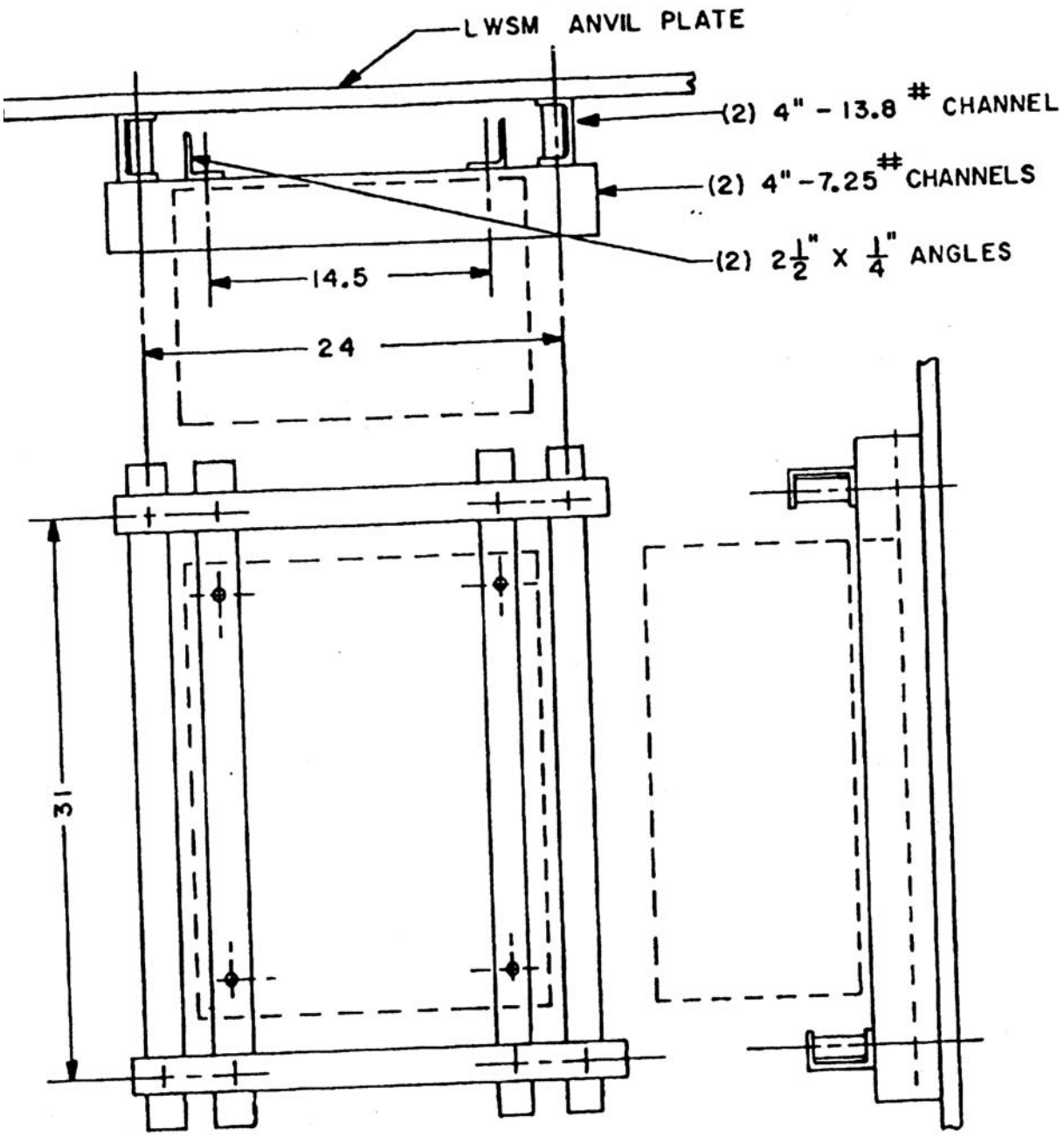


FIGURE 2. Standard mounting fixtures for shock testing of type AQB-A1602 circuit breakers.

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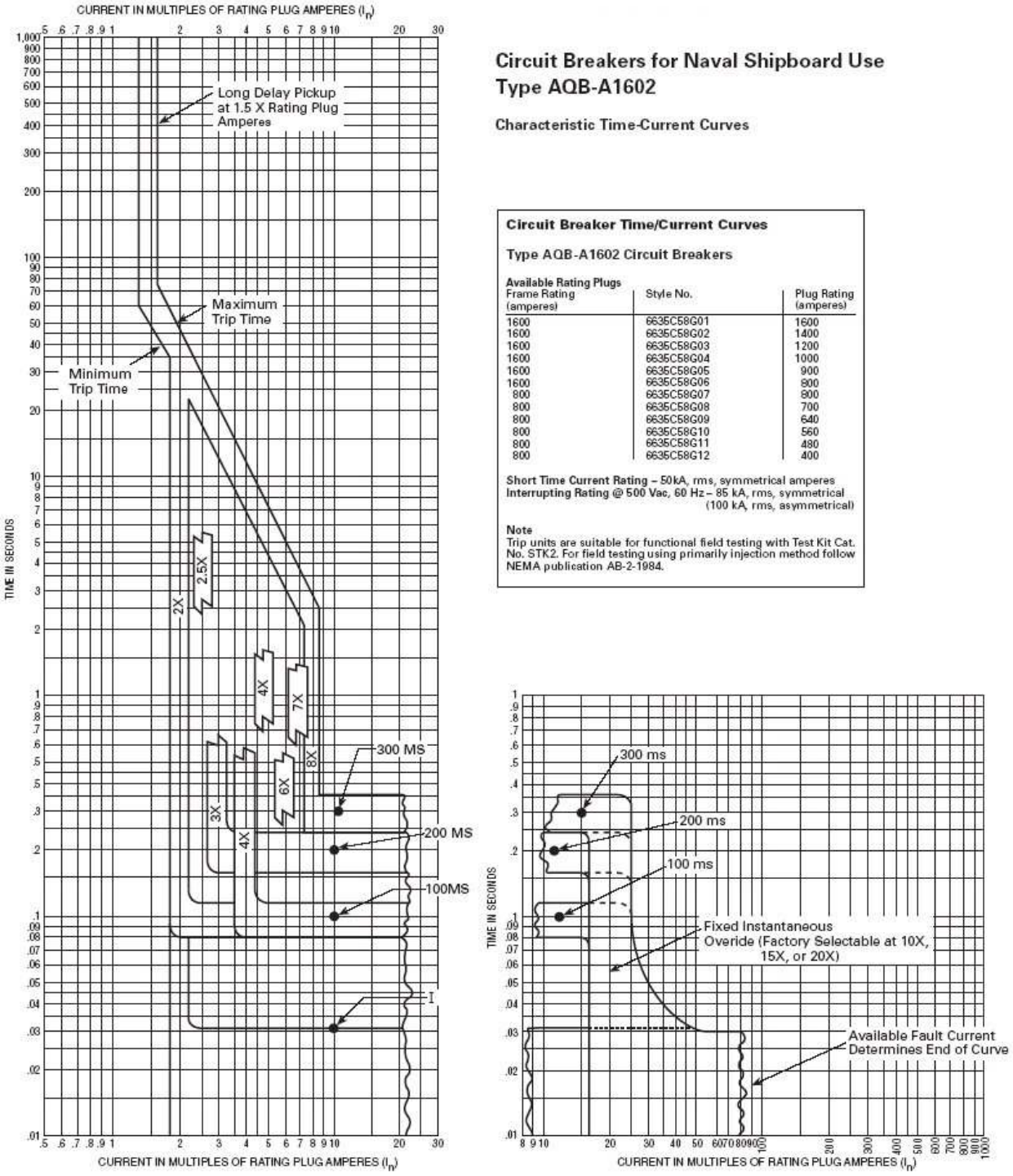


FIGURE 3. Circuit breaker type AQB-A1602 time-current response.

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CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Preparing Activity:
Navy – SH
(Project 5925-N185-015)

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